Evaluating Two-Input Transfer Function Analysis of Cerebral Hemodynamics

Known-truth simulation framework and a 250-family proof of concept

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August 2026

How this presentation is being used

PURPOSE

This presentation is organized like a developing paper rather than a short conference talk. The early sections establish the signal-processing and physiological reasoning step by step. The later sections define the simulation, show how estimates are compared with known truth, and then introduce the first proof-of-concept results.

The goal is not to make the two-input model look better by default. The goal is to identify the conditions in which jointly estimating MAP and $P_{\mathrm{ET}}CO_2$ recovers their separate contributions to CBFV more accurately than treating each input alone.

SECTION 1

Introduction and Motivation

CBFV is influenced by more than arterial pressure. That creates a reason to consider two inputs, but it does not guarantee that a larger model will be more accurate.

Cerebral blood flow is regulated despite changing pressure

INTRODUCTION AND MOTIVATION

The brain requires continuous blood flow even though systemic arterial pressure changes from moment to moment. Cerebral autoregulation describes part of the vascular response that limits how strongly these pressure fluctuations alter cerebral perfusion. Dynamic cerebral autoregulation focuses on changes occurring over seconds to minutes and is commonly studied by treating MAP as an input and CBFV as an output.

Transfer function analysis is useful because it describes this relationship at each frequency. Gain represents the size of the response and phase represents its timing. These measures describe a dynamic relationship; they do not prove that one signal is the only physiological cause of the other.

$P_{\mathrm{ET}}CO_2$ creates a second measured influence on CBFV

INTRODUCTION AND MOTIVATION

Carbon dioxide changes cerebrovascular tone and can influence CBFV in frequency ranges that overlap with the ranges used for autoregulation analysis. End-tidal carbon dioxide is not identical to arterial carbon dioxide, but it provides a continuous practical measurement during resting recordings.

If MAP and $P_{\mathrm{ET}}CO_2$ share fluctuations, a MAP-only transfer function can contain variation associated with both pathways. The same issue can occur in the reverse direction for a $P_{\mathrm{ET}}CO_2$-only estimate. Separate SISO estimates remain valid pairwise descriptions, but they should not automatically be interpreted as fully separated physiological pathways.

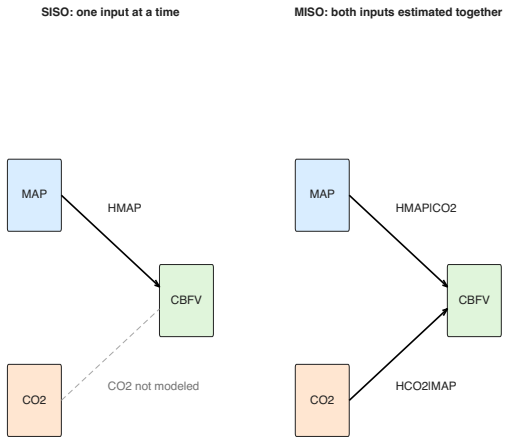
The study asks a conditional question

INTRODUCTION AND MOTIVATION

A two-input model provides a mathematical way to separate shared information, but it must estimate two pathway coefficients from the same finite recording. When the inputs are difficult to distinguish, the joint spectral system can become sensitive to small estimation errors.

The useful question is therefore: *under which combinations of input relationship, pathway strength, recording quality, and estimator settings does MISO recover the known pathways more accurately than separate SISO models?* Human recordings cannot answer this accuracy question by themselves because the true physiological transfer functions are not directly observable.

SISO and MISO answer related but different questions



Conceptual method diagram generated by the MATLAB project; it does not present numerical evidence.

SECTION 2

From Physiological Signals to Spectral Estimates

Transfer functions are estimated from auto-spectra and cross-spectra. The foundation begins with how a finite time signal becomes a stable spectral estimate.

A transfer function maps an input to an output

SIGNAL-PROCESSING BACKGROUND

If $x(t)$ is an input and $y(t)$ is an output, a simple static analogy is $y = mx$: the slope m describes how strongly the input is represented in the output. A transfer function plays a similar role but can have a different value at every frequency and can also describe timing.

$$Y(f) = H(f)X(f)$$

The magnitude $|H(f)|$ is gain, while the angle $\arg H(f)$ is phase. Together they describe the size and timing of the output response at frequency f.

The frequency domain reorganizes the same signal

SIGNAL-PROCESSING BACKGROUND

MAP, $P_{\mathrm{ET}}CO_2$, and CBFV are first recorded in the time domain. Each sample describes the signal at one time, which is useful for seeing trends, artifacts, noise, and timing relationships. Several oscillations can overlap in a single time trace, however, making their individual contributions hard to see.

The Fourier transform reorganizes that finite record into discrete frequency values. Slow oscillations appear at low frequencies and faster oscillations at higher frequencies. Each Fourier value is generally complex: its magnitude describes the strength of that component and its angle describes phase relative to a reference sinusoid.

Frequency spectra and power spectra retain different information

SIGNAL-PROCESSING BACKGROUND

The complex frequency spectrum retains both magnitude and phase. A power spectrum is based on squared magnitude and describes how signal power is distributed across frequency; it is real and nonnegative. Cross-spectra combine two signals and preserve their relative phase information.

Directly dividing two single-record Fourier spectra gives useful intuition,

$$\widehat{H}(f) = \frac{Y(f)}{X(f)},$$

but this ratio can become unstable when $X(f)$ is small or when one realization of noise dominates a frequency. TFA therefore uses averaged spectral estimates rather than one direct FFT division.

Welch's method creates repeated estimates inside one record

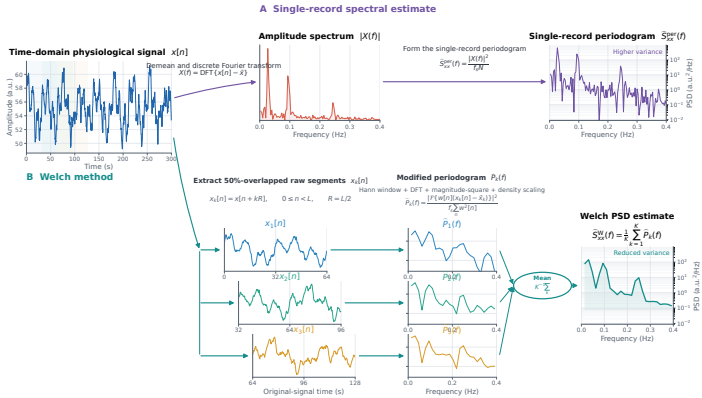
SIGNAL-PROCESSING BACKGROUND

Welch's method divides the original record into shorter, often overlapping segments. Every segment contains the actual samples from its corresponding time interval; overlapping segments share some of those samples rather than creating different versions of the signal.

Each segment is multiplied by a window such as a Hann window, transformed, and converted into a segment-level spectral estimate. Because all segments use the same frequency grid, their periodograms or cross-periodograms can be averaged frequency by frequency. This reduces variance but creates a tradeoff: shorter windows provide more averages, whereas longer windows provide finer frequency resolution.

Two routes from one time signal to a spectral estimate

SIGNAL-PROCESSING BACKGROUND



The lower route shows overlapping samples from one signal, segment-level modified periodograms, and their frequency-by-frequency average.

Auto-spectra describe power within one signal

SIGNAL-PROCESSING BACKGROUND

For segment m, let $X_m(f)$ be the Fourier transform of the windowed signal. The segment power estimate contains the product of the complex value and its own complex conjugate. Welch's estimate averages these products across the M available segments.

$$\widehat{S}_{xx}(f) = \frac{1}{M}\sum_{m=1}^{M} X_m^*(f)X_m(f) = \frac{1}{M}\sum_{m=1}^{M} |X_m(f)|^2$$

The implementation also includes the sampling-frequency and window-energy normalization needed for a correctly scaled power spectral density.

Cross-spectra retain the relationship between two signals

SIGNAL-PROCESSING BACKGROUND

For signals X and Y, the cross-spectrum uses the conjugate of the first signal and the Fourier value of the second signal. Its magnitude describes shared frequency content, while its angle contains the relative phase.

$$\widehat{S}_{xy}(f) = \frac{1}{M} \sum_{m=1}^{M} X_m^*(f) Y_m(f)$$

The ordering must be stated because reversing the two signals conjugates the cross-spectrum and changes the sign of phase.

SECTION 3

Conventional SISO Transfer-Function Analysis

The SISO estimator describes one selected input-output relationship at a time.

A dynamic time-domain response becomes multiplication by frequency

SISO TRANSFER-FUNCTION ANALYSIS

The output at one time can depend on the current and previous values of the input. The impulse response $h(t)$ describes how those values contribute across time, including delayed and distributed effects.

$$y(t) = (h * x)(t),$$
$$Y(f) = H(f)X(f).$$

The Fourier transform changes convolution into a frequency-by-frequency multiplication. This is why transfer function analysis can estimate the dynamic relationship separately at each frequency.

The SISO estimator uses averaged spectral quantities

SISO TRANSFER-FUNCTION ANALYSIS

Rather than dividing one output FFT by one input FFT, the estimator divides the averaged input-output cross-spectrum by the averaged input auto-spectrum.

$$\widehat{H}_x^{\mathrm{SISO}}(f) = \frac{\widehat{S}_{xy}(f)}{\widehat{S}_{xx}(f)}$$

The numerator describes how the input and output vary together. The denominator scales that relationship by the input power available at the same frequency. Hats indicate quantities estimated from a finite observation.

Coherence describes consistency, not physiological truth

SISO TRANSFER-FUNCTION ANALYSIS

Magnitude-squared coherence compares the input-output cross-spectrum with the power present in both signals.

$$\gamma_{xy}^2(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)}, \qquad 0 \leq \gamma_{xy}^2(f) \leq 1$$

High coherence means that a consistent linear pairwise relationship is present at that frequency. It does not prove causality, and it does not show that a pairwise SISO estimate equals the underlying MAP or $P_{\mathrm{ET}}CO_2$ pathway after the other measured input is considered.

A SISO estimate is a pairwise description

SISO TRANSFER-FUNCTION ANALYSIS

The MAP SISO transfer function answers how MAP and CBFV are related when MAP is the only modeled input. The $P_{\mathrm{ET}}CO_2$ SISO transfer function answers the corresponding pairwise question for $P_{\mathrm{ET}}CO_2$ and CBFV.

Neither calculation is mathematically wrong. The limitation appears when the result is interpreted as a uniquely separated pathway even though the other measured input is related to both the selected input and the output. This distinction motivates the MISO derivation rather than an assumption that a larger model must be better.

SECTION 4

Why a SISO Estimate Can Contain the Second Pathway

The shared-input term can be derived directly from the true two-input system.

Begin with a true two-input linear system

WHY A SECOND INPUT MATTERS

Let M represent MAP, C represent $P_{\mathrm{ET}}CO_2$, and Y represent CBFV. Each input has its own true transfer function, while E contains output variation not explained by the two measured inputs.

$$Y(f) = H_M(f)M(f) + H_C(f)C(f) + E(f)$$

For the spectral derivation, the unexplained term is assumed to be uncorrelated with both measured inputs. The known-truth simulation later adds finite-record variation and measurement noise while retaining known H_M and H_C.

Relating MAP to the output keeps both pathway terms

WHY A SECOND INPUT MATTERS

Multiply the two-input system by $M^*(f)$ and take the expectation. The MAP-to-output cross-spectrum contains the MAP pathway multiplied by MAP power, but it also contains a second term whenever MAP and $P_{\mathrm{ET}}CO_2$ have a nonzero cross-spectrum.

$$S_{MY}(f) = H_M(f)S_{MM}(f) + H_C(f)S_{MC}(f)$$

The second term is not noise. It represents the true $P_{\mathrm{ET}}CO_2$ pathway carried into a pairwise MAP-output relationship through the relationship between the two inputs.

The MAP SISO estimate can include a $P_{\mathrm{ET}}CO_2$ term

WHY A SECOND INPUT MATTERS

Substituting the two-input cross-spectrum into the SISO estimator gives a direct expression for what MAP-only TFA estimates.

$$H_M^{\mathrm{SISO}}(f) = H_M(f) + H_C(f)\frac{S_{MC}(f)}{S_{MM}(f)}$$

The first term is the MAP pathway. The second depends on the $P_{\mathrm{ET}}CO_2$ pathway and on the complex relationship between the two inputs. Its magnitude and phase can change across frequency, so it cannot generally be removed with one scalar correction.

The reverse occurs for the $P_{\mathrm{ET}}CO_2$ SISO estimate

WHY A SECOND INPUT MATTERS

Repeating the same derivation with $P_{\mathrm{ET}}CO_2$ as the selected input produces the corresponding shared-input term.

$$H_C^{\mathrm{SISO}}(f) = H_C(f) + H_M(f)\frac{S_{CM}(f)}{S_{CC}(f)}$$

If $S_{MC}(f) = 0$, this specific mixing term disappears at that frequency. This special case explains why MISO should not be expected to improve every record equally: when the inputs are nearly independent, there may be little SISO mixing to remove.

SECTION 5

The Joint MISO Model

MISO solves both pathway coefficients together using the complete two-input spectral matrix.

The two pathway estimates are solved together

THE MISO MODEL

At each frequency, the diagonal terms contain MAP and $P_{\mathrm{ET}}CO_2$ power, the off-diagonal terms contain their cross-spectrum, and the right-side vector connects each input to CBFV.

$$\begin{bmatrix} S_{MM} & S_{MC} \\ S_{CM} & S_{CC} \end{bmatrix} \begin{bmatrix} H_M \\ H_C \end{bmatrix} = \begin{bmatrix} S_{MY} \\ S_{CY} \end{bmatrix}$$

The system uses both unique and shared input information. The resulting pathway coefficients are conditional estimates: each describes one input-output relationship while accounting for the other measured input.

The implementation uses a linear-system solve

THE MISO MODEL

The compact estimator is written with the input spectral matrix $\widehat{S}_{xx}$ and input-output cross-spectrum vector $\widehat{S}_{xy}$.

$$\widehat{H}(f) = \widehat{S}_{xx}(f) \backslash \widehat{S}_{xy}(f)$$

The backslash denotes solving the two-by-two linear system. The code does not form an explicit matrix inverse. It also symmetrizes the finite-sample spectral matrix and checks input power and conditioning before selecting a solution.

The core MATLAB solve stays close to the mathematics

CODE IMPLEMENTATION

```
inputSpectralMatrix = ...
    (inputSpectralMatrix + inputSpectralMatrix') / 2;

inputScale = sqrt(real(diag(inputSpectralMatrix)));
scaleMatrix = diag(inputScale);

normalizedMatrix = ...
    scaleMatrix \ inputSpectralMatrix / scaleMatrix;
normalizedCrossSpectrum = ...
    scaleMatrix \ inputOutputSpectrum;

directCoefficients = ...
    inputSpectralMatrix \ inputOutputSpectrum;
```

Condensed from `src/analysis/solveMisoSpectralSystem.m`.

Identifiability means that the two pathways can be distinguished

THE MISO MODEL

The joint model requires some MAP variation that cannot be predicted from $P_{\mathrm{ET}}CO_2$ and some $P_{\mathrm{ET}}CO_2$ variation that cannot be predicted from MAP. For two inputs, the determinant makes this requirement visible.

$$D(f) = S_{MM}(f)S_{CC}(f) - |S_{MC}(f)|^2$$

When the determinant approaches zero, the two equations contain nearly redundant information. Small spectral estimation errors can then produce large changes in the separated pathway coefficients.

Input coherence connects the physiology to matrix conditioning

THE MISO MODEL

Magnitude-squared input coherence can be substituted into the determinant:

$$D(f) = S_{MM}(f)S_{CC}(f)\left[1 - \gamma^2_{MC}(f)\right].$$

As input coherence approaches one, the normalized determinant approaches zero. This does not mean that any shared input variation invalidates MISO. It means that the benefit of removing SISO mixing must be weighed against the declining amount of independent information available to separate the two coefficients.

Scaling distinguishes units from the input relationship

THE MISO MODEL

MAP and $P_{\mathrm{ET}}\,CO_2$ have different units and may have very different power. Before ridge sensitivity is evaluated, the implementation scales each input to unit spectral power. The normalized matrix is

$$\widetilde{\mathrm{S}}_{xx}(f) = \begin{bmatrix} 1 & \rho(f) \\ \rho^*(f) & 1 \end{bmatrix}, \qquad |\rho(f)|^2 = \gamma^2_{MC}(f).$$

Condition numbers calculated from this matrix describe the relationship between the two inputs rather than being dominated only by their physical units. Ridge regularization is added in this normalized space during sensitivity analysis.

SECTION 6

Input Coherence, PSD Overlap, and Excitation

These ideas are related, but each describes a different part of the information available to a two-input model.

Input coherence describes shared fluctuations

INPUT RELATIONSHIPS AND EXCITATION

MAP-$P_{\mathrm{ET}}CO_2$ coherence measures how consistently the two inputs are linearly related at each frequency. High coherence means that their fluctuations are strongly connected there; low coherence means that more of the observed variation is independent.

Coherence concerns the relationship between the two inputs. It does not say whether either input contains enough power to estimate its pathway precisely, and it is not interchangeable with the amount of overlap between their power spectral density shapes.

PSD overlap describes shared frequency support

INPUT RELATIONSHIPS AND EXCITATION

PSD-shape overlap asks whether MAP and $P_{\mathrm{ET}}CO_2$ contain power in similar frequency ranges. Two inputs can occupy the same broad frequencies without having strongly coherent sample-by-sample fluctuations. Conversely, a pathway cannot be estimated where its input has almost no measurable power, even if a nominal relationship exists elsewhere.

The simulation therefore varies a spectral-similarity control separately from target coherence. Its realized PSD-shape overlap is measured from a long clean record and later grouped into prespecified fixed intervals for descriptive operating maps.

Excitation separates input strength from pathway strength

INPUT RELATIONSHIPS AND EXCITATION

Input excitation describes how strongly an input fluctuates in the analyzed frequency range. Pathway gain describes how strongly CBFV responds to those fluctuations. A weak $P_{\mathrm{ET}}CO_2$ input and a weak $P_{\mathrm{ET}}CO_2$-to-CBFV pathway can both produce a small output contribution, but they are different physiological and estimation conditions.

The simulation varies the $P_{\mathrm{ET}}CO_2$-to-MAP fluctuation standard-deviation ratio and the $P_{\mathrm{ET}}CO_2$-to-MAP pathway band-gain ratio separately so that their joint effect can be examined rather than compressed into one control.

The expected MISO advantage lies between two extremes

INPUT RELATIONSHIPS AND EXCITATION

With nearly independent inputs, SISO contains little shared-input mixing, so the additional MISO coefficient may provide limited benefit. With moderate shared variation and remaining independent information, MISO may separate the pathways more accurately. With extremely high coherence, the input spectral system can become poorly conditioned and the separated solution can become unstable.

This is a hypothesis about an operating region, not a conclusion. The known-truth simulation is needed because a model that looks plausible or has higher in-sample multiple coherence is not necessarily closer to the true pathways.

SECTION 7

Known-Truth Simulation Design

The simulation separates fixed physiological families from repeated observation conditions and estimator settings, then compares every estimate with known transfer functions.

A family, observation, and estimator setting are different units

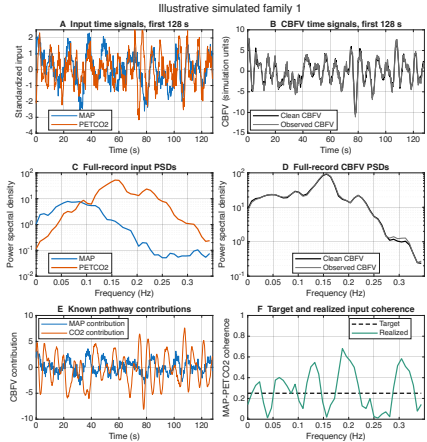
SIMULATION DESIGN

Unit	Meaning in the simulation
Family	One fixed input structure and pair of true MAP and $P_{\mathrm{ET}}CO_2$ pathways. The family is the independent statistical unit.
Observation	A different way of recording the same family: duration, output noise, input measurement noise, or timing alignment may change while truth remains fixed.
Estimator setting	A different analysis of the same observation, such as Welch window length, overlap, or ridge value.

Welch segments, frequency bins, and repeated observations are not counted as independent families.

One simulated family keeps the pathway truth visible

SIMULATION DESIGN



Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Five intrinsic controls define the proof-of-concept families

SIMULATION DESIGN

Assigned family factor	Five configured values
Target MAP-$P_{ET}CO_2$ coherence	0.05, 0.25, 0.50, 0.75, 0.95
Spectral-similarity control	0, 0.25, 0.50, 0.75, 1
$P_{ET}CO_2$/MAP fluctuation SD ratio	0.05, 0.25, 0.50, 1, 2
$P_{ET}CO_2$/MAP pathway band-gain ratio	0.10, 0.50, 1, 2, 4
True $P_{ET}CO_2$ pathway delay	0, 2.5, 5, 7.5, 10 s

MAP and $P_{ET}CO_2$ pathway time constants remain continuously sampled nuisance properties over 1–4 s and 2–8 s. They are not forced into categories only to make the figures look uniform.

Pairwise balance replaces an unnecessary full factorial

SIMULATION DESIGN

A complete five-factor design at five levels would require

$$5^5 = 3{,}125 \text{ families.}$$

The proof design instead uses a 25-family base block in which each factor level appears five times and every pair of factors contains all 5×5 level combinations once. Ten independently generated blocks give

$$25 \text{ families per block} \times 10 \text{ blocks} = 250 \text{ families.}$$

Therefore each assigned level has 50 families and each assigned pairwise cell has 10. The design protects planned one-factor and two-factor summaries without claiming to cover all 3,125 five-factor combinations.

Assigned factors and realized characteristics must remain distinct

SIMULATION DESIGN

Assigned factors are selected before signal generation and are displayed at their exact configured values. Realized characteristics are measured from the longest clean generated record. They can differ slightly from their targets and are continuous rather than five-valued by definition.

For presentation operating maps, realized input coherence, PSD-shape overlap, and $P_{\mathrm{ET}}CO_2$ contribution-power share use five fixed intervals:

$$[0, 0.2),\ [0.2, 0.4),\ [0.4, 0.6),\ [0.6, 0.8),\ [0.8, 1].$$

These boundaries were specified before the 250-family results were inspected and were not moved afterward merely to fill heatmap cells.

Every family receives the same paired observation plan

SIMULATION DESIGN

Experiment	Levels applied to every family
Duration by CBFV SNR	Seven durations (256–896 s) crossed with seven output-SNR levels (∞ to 0 dB): 49 observations
MAP measurement noise	Seven input-SNR levels at the reference observation
$P_{\mathrm{ET}}CO_2$ measurement noise	Seven input-SNR levels at the reference observation
$P_{\mathrm{ET}}CO_2$ timing misalignment	$-6, -3, 0, 3, 6$ s at the reference observation

The reference observation is 300 s with 15-dB CBFV output SNR, no added input measurement noise, and no timing misalignment. The plan contains 68 primary observations per family, producing $250 \times 68 = 17{,}000$ primary analyses.

Estimator sensitivity is repeated within each family

SIMULATION DESIGN

The primary estimator uses a 128-s Hann window, 50% overlap, and the common three-point smoothing kernel [0.25, 0.50, 0.25]. At least three complete Welch windows are required.

Sensitivity analyses evaluate ridge values 0.001, 0.01, and 0.1; Welch windows of 64, 75, 100, 128, and 150 s; overlaps of 25%, 50%, and 75%; and the full duration-by-window grid. This adds 750 ridge analyses and 10,750 estimator-setting analyses to the 250-family run.

Because these are repeated analyses of the same underlying families, they increase the number of computations without increasing the independent sample size beyond 250.

The named profile makes the intended analysis explicit

CODE IMPLEMENTATION

```
% "proof250": presentation proof of concept
simulationProfile = lower("proof250");

paperSettings = approachPaperSettings( ...
    dataFolder, simulationOutputFolder, ...
    simulationProfile, analysisSettings);

paperSettings.steps.runEmpiricalAnalysis = false;
paperSettings.steps.runKnownTruthSimulation = true;
paperSettings.steps.createTables = true;
paperSettings.steps.createFigures = true;

runResults = runApproachPaper(paperSettings);
```

Condensed from `main.m`. The profile defines family count, factor levels, observation plan, fixed proof bins, and estimator sensitivities.

SECTION 8

How the Estimates Are Compared With Truth

The primary outcome compares the complete complex transfer function, while gain and phase errors explain which part of the estimate changed.

Complex error combines magnitude and phase disagreement

KNOWN-TRUTH OUTCOMES

For an estimated pathway $\widehat{H}(f)$ and its known truth $H(f)$, the code averages squared complex distance over the analyzed frequency range and normalizes by the mean energy of the true transfer function:

$$E_{\text{complex}} = \frac{\text{mean}_f \left| \widehat{H}(f) - H(f) \right|^2}{\text{mean}_f \, |H(f)|^2}.$$

This metric penalizes disagreement in either real or imaginary components and therefore retains both gain and phase information. Separate absolute gain and wrapped phase errors are also calculated for interpretation.

Phase error is evaluated only where the true pathway is defined

KNOWN-TRUTH OUTCOMES

Phase becomes unstable or physically uninformative where true pathway gain is nearly zero. The implementation therefore evaluates wrapped phase error only at frequencies where true gain exceeds 5% of its root-mean-square gain.

$$e_{\phi}(f) = \angle\left[\exp\left(i\{\angle\widehat{H}(f) - \angle H(f)\}\right)\right].$$

This returns the principal angular difference rather than treating a 2π representation change as a large physiological error.

Model advantage expresses the relative error on a log scale

KNOWN-TRUTH OUTCOMES

For the same family and observation, SISO and MISO error are converted into a model-advantage value:

$$A_{\mathrm{model}} = \log_{10}\left(\frac{E_{\mathrm{SISO}} + \epsilon}{E_{\mathrm{MISO}} + \epsilon}\right), \qquad \epsilon = 10^{-12}.$$

Positive values indicate lower MISO error, zero indicates equal error, and negative values indicate lower SISO error. A value of 0.30 corresponds to approximately a twofold error ratio; a value of 1 corresponds to a tenfold ratio. Absolute errors remain necessary because a favorable ratio can coexist with poor performance by both models.

Aggregation preserves the family as the independent unit

KNOWN-TRUTH OUTCOMES

Each displayed group or heatmap cell is summarized across families, not across Welch segments or frequency bins treated as independent repetitions. Exported source data retain the mean, standard deviation, valid family count, total group count, standard error, confidence interval, and probability values when inference is performed.

The presentation display threshold is five valid families. Cells below this threshold remain gray and receive no mean color. Five families permit a descriptive calculation; they do not by themselves provide strong publication evidence. Frequency-resolved plots keep the native Welch frequency bins rather than rebinned frequencies.

SECTION 9

Audited 250-Family Proof-of-Concept Results

The run verifies the design and demonstrates the expected MISO accuracy–stability tradeoff. These results guide the final paper design rather than replace it.

The proof-of-concept run completed as designed

SIMULATION AUDIT

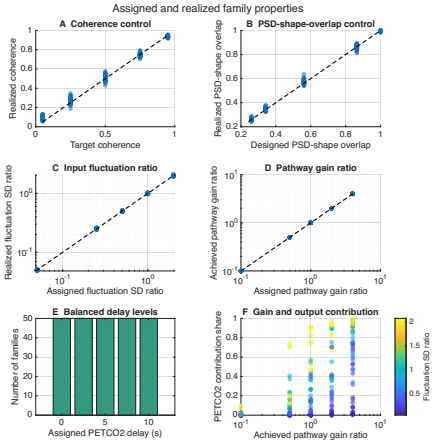
Completed quantity	Count
Independent simulated families	250
Complete 25-family balanced blocks	10
Primary observation analyses	17,000
Ridge analyses	750
Estimator-setting analyses	10,750
Audited metric-condition support checks	748
Numerical runtime	21.3 min

All 14 core execution checks passed. All 25 marginal level checks and all 250 assigned pairwise-cell checks passed. No audited whole-range metric value was silently discarded as nonfinite.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Assigned controls were reproduced by the generator

GENERATOR VALIDATION



Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Balanced checkpoints support using 250 families for this deck

SIMULATION AUDIT

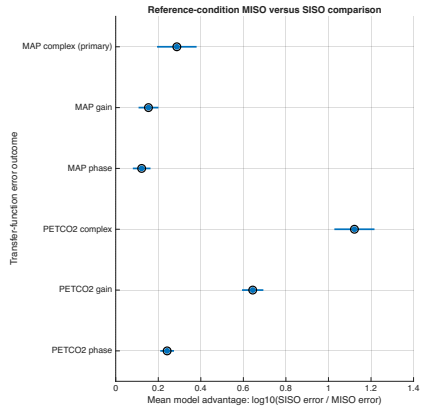
The design contains natural complete-block checkpoints at 125, 200, and 250 families. Across the six reference-condition complex, gain, and phase model-advantage outcomes, the largest absolute difference from the final 250-family mean was 0.0339 at 125 families and 0.0169 at 200 families.

These are descriptive stability checks rather than additional hypothesis tests. They show that the main proof-of-concept averages were not being driven by the last few families, while still leaving the larger 1,000-family profile necessary for stronger cell support and final publication-level modeling.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

At the reference condition, MISO had lower mean error

REFERENCE-CONDITION RESULT



Means and 95% confidence intervals across 250 families. Black rings mark the reported comparison after the stated adjustment family. Model advantage is $\log_{10}(E_{\mathrm{SISO}} / E_{\mathrm{MISO}})$.

The reference result was larger for the $P_{\mathrm{ET}}CO_2$ pathway

REFERENCE-CONDITION RESULT

At 300 s, 15-dB CBFV output SNR, and no input noise or timing shift, mean MAP complex model advantage was 0.287 (95% CI 0.194 to 0.380). This corresponds to an approximately 1.94-fold geometric SISO-to-MISO error ratio.

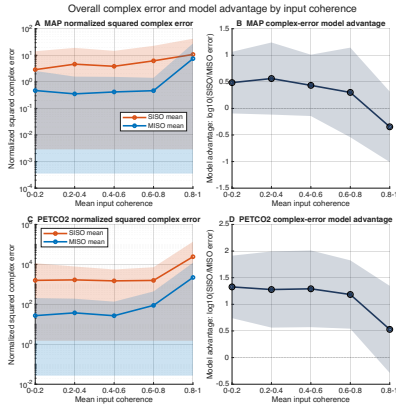
Mean $P_{\mathrm{ET}}CO_2$ complex model advantage was 1.122 (95% CI 1.028 to 1.216), corresponding to an approximately 13.25-fold geometric error ratio. Gain and phase comparisons were also positive on average for both pathways.

This establishes a proof-of-concept average difference, not a claim that MISO is better in every family. The next analyses examine where that average changes.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Very high input coherence changes the MAP tradeoff

INPUT IDENTIFIABILITY RESULT



Families are grouped by prespecified intervals of realized mean input coherence. Curves show across-family means; shaded regions show SD.

The highest-coherence group illustrates the stability limit

INPUT IDENTIFIABILITY RESULT

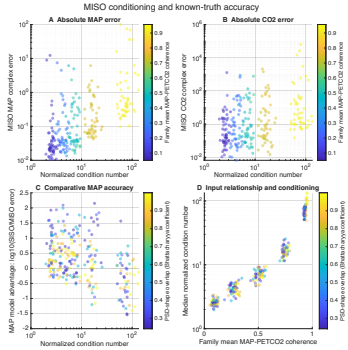
MAP model advantage remained positive through the 0.6–0.8 realized-coherence group, but the 0.8–1.0 group had a mean advantage of −0.346 (95% CI −0.534 to −0.157). Its geometric SISO-to-MISO error ratio was 0.45, indicating lower MAP SISO error on average in that group.

The $P_{\mathrm{ET}}CO_2$ advantage also decreased at the highest coherence but remained positive on average: 0.529 (95% CI 0.297 to 0.762). This difference between pathways is important. Poor identifiability does not have to affect both true pathways in the same way because their excitation and pathway gains also differ.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Condition number connects coherence with known-truth error

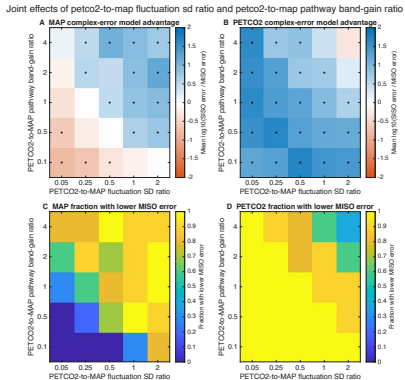
INPUT IDENTIFIABILITY RESULT



Each point represents one family at the reference observation. Conditioning is calculated after scaling the inputs to unit spectral power.

Excitation and pathway gain must be interpreted together

PATHWAY EXCITATION RESULT



Axes use exact assigned values; each 5×5 pairwise cell contains 10 families. Black dots mark adjusted $P < 0.05$.

Weak $P_{\mathrm{ET}}CO_2$ influence can limit the MAP benefit

PATHWAY EXCITATION RESULT

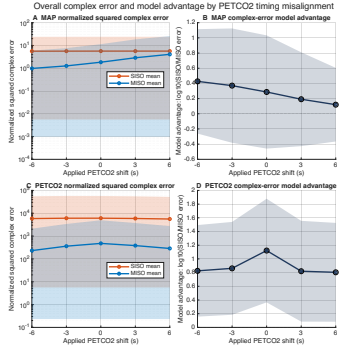
The joint assigned-factor map separates the strength of $P_{\mathrm{ET}}CO_2$ fluctuations from the strength of its pathway to CBFV. MAP MISO advantage is weakest, and can be negative, when both the $P_{\mathrm{ET}}CO_2$ input and its pathway are very weak. In that region there is little second-pathway influence for MISO to separate, so an additional coefficient can add finite-record variability without removing much SISO mixing.

As either the $P_{\mathrm{ET}}CO_2$ fluctuations or pathway gain becomes stronger, MAP MISO advantage generally increases. The $P_{\mathrm{ET}}CO_2$ pathway shows a broader positive MISO advantage, but its exact magnitude also changes across the joint design. This supports interpreting excitation and pathway strength as separate controls rather than one combined label.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Timing alignment changes the two pathways differently

OBSERVATION-QUALITY RESULT



Each timing shift is applied to the same 250 families at the reference duration and output SNR. Curves show across-family means; shaded regions show SD.

The timing result is directional rather than a single penalty

OBSERVATION-QUALITY RESULT

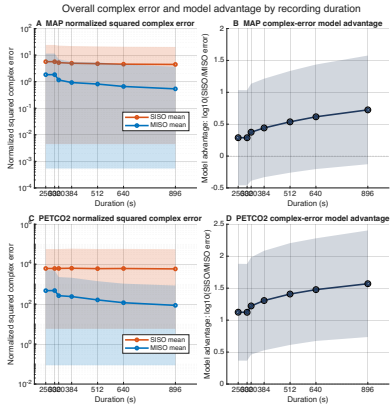
The $P_{\mathrm{ET}}CO_2$ complex model advantage was largest with no applied timing shift (1.122) and decreased to approximately 0.80–0.86 after shifts of three or six seconds in either direction. MAP advantage changed asymmetrically: it was 0.287 at zero shift, 0.120 at $+6$ s, and 0.427 at -6 s.

This asymmetry is consistent with timing error interacting with the families' true $P_{\mathrm{ET}}CO_2$ delays rather than acting as one uniform amount of noise. A recording's alignment choice can therefore change which complex pathway is easier to recover and should be treated as part of the model specification.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Longer records generally strengthen the advantage

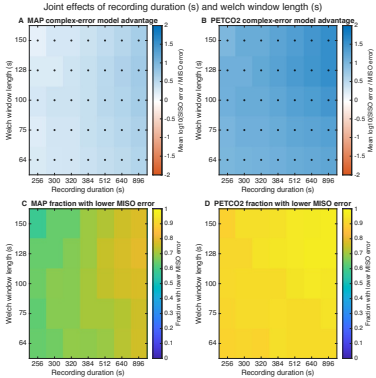
FINITE-RECORD RESULT



Durations use exact configured values. All curves use 15-dB CBFV output SNR and the primary 128-s, 50%-overlap Welch estimator.

Duration and Welch window length form one estimator tradeoff

FINITE-RECORD RESULT



Axes use exact configured values. Each cell summarizes the same 250 families; black dots mark adjusted $P < 0.05$.

The interaction map supports an intuitive interpretation

FINITE-RECORD RESULT

Longer recordings generally increase the amount of information available to both models and increase the fraction of families in which MISO has lower complex error. Window length modifies this result because it controls the balance between frequency resolution and the number of segment-level estimates that can be averaged.

The $P_{\mathrm{ET}}CO_2$ pathway shows a broad MISO advantage across the tested duration-window grid. The MAP pathway shows more gradual improvement with duration and greater dependence on window choice. This difference reinforces that estimator settings should not be selected only from one pathway's behavior.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

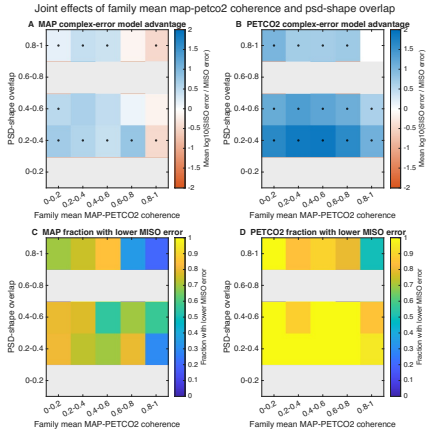
SECTION 10

What the Proof of Concept Establishes–and What It Does Not

The run supports the workflow and the main tradeoff, while also identifying design decisions that must be made before the final paper simulation.

Realized bins reveal where the current design is sparse

PROOF-OF-CONCEPT LIMITATIONS



Gray cells contain fewer than five valid families. The fixed realized-value boundaries were not moved after seeing occupancy.

The gray cells are information about support, not missing results

PROOF-OF-CONCEPT LIMITATIONS

Realized coherence occupied the five bins almost evenly: 51, 49, 50, 50, and 50 families. Realized PSD-shape overlap was uneven: 0, 100, 45, 5, and 100. Across the three audited two-dimensional realized-characteristic grids, 43 of 75 cells met the five-family display threshold and 13 were empty.

The unsupported cells remain gray because the proof design did not generate enough families there. Filling them by changing bin boundaries after seeing the data would make the figure look more complete without adding information. These maps are therefore descriptive and should not be treated as precise boundaries of an operating region.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

The 1,000-family run is a planned extension, not a simple rerun

TRANSITION TO THE FINAL PAPER

Repeating the 25-family balanced block 40 times would provide 1,000 families, 40 families per assigned pairwise cell, and greater support for continuous models or finer descriptive grouping. Before that run, the following choices should be specified prospectively:

- final continuous-bin boundaries and minimum cell size;
- which interactions are primary, secondary, or supplemental;
- whether spectral-similarity coverage should be expanded;
- the primary endpoint and multiplicity families; and
- which family-level continuous regression models will support formal claims.

The larger sample should increase evidence, not replace the design decisions revealed by the proof of concept.

The current conclusion is methodological and conditional

PROOF-OF-CONCEPT SYNTHESIS

The 250-family run shows that MISO can recover both known pathways more accurately on average under the reference observation, with an especially large average advantage for the $P_{\mathrm{ET}}CO_2$ pathway. It also shows why that conclusion cannot be reduced to "MISO is better." MAP accuracy can favor SISO when the inputs are extremely coherent, and MISO benefit changes with excitation, pathway gain, timing, record duration, and estimator settings.

The next scientific step is to finalize the paper-scale design using these audit findings, run the 1,000-family profile, and only then place the cognitively normal human recordings within the supported simulated operating space.

REVIEW STOPPING POINT

The method and proof-of-concept results are now connected.

The next review should focus on whether the narrative matches the intended paper direction, whether the selected figures communicate the main tradeoffs, and which proof findings should determine the final 1,000-family simulation specification.